Dominic Roberts, Ph.D.

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Full-stack AI professional and machine learning researcher, with >4.5 years of industry experience.

Work Experience

Senior Applied Scientist @ Microsoft 01/26 – present
Redmond, WA ***(1 month)***

Responsibilities:

- Develop evaluation frameworks for LLM-based voice agents when subject to echo, reverb and distortion
- Devise metrics for evaluating audio quality in Microsoft Teams and Azure Communication Services calls

Machine Learning Scientist @ steg.ai 04/24 – 12/25
Bellevue, WA ***(1 year, 7 months)***

Responsibilities:

- Develop multi-modal CNN- and transformer-based models for imperceptibly watermarking audio, images and videos, exploring tradeoffs between watermark imperceptibility and payload recovery rate
- Productionize models on Amazon EC2 instances and optimize for memory and latency via custom inference logic
- Expand the company's visible and invisible watermarking solutions to new types of digital assets
- Review and benchmark state-of-the-art steganography methods

Key achievements:

- Increased robustness of invisible watermarks to image operations such as JPEG compression, cropping and perspective warping by 80% without impacting watermark imperceptibility
- Created Python-based software for high-quality and rapid PDF watermarking, key to signing a commercial contract with a popular media company (name subject to NDA)

Applied Scientist II @ Amazon, Physical Stores org 07/21 – 03/24
Seattle, WA ***(2 years, 8 months)***

Responsibilities:

- Ideate, train, productionize and oversee continuous improvement of retrieval, ranking and classification components based on vision-language models in a new customer experience in Amazon Fresh Stores
- Optimize business metrics related to Amazon's deep learning-based Just Walk Out technology (based on action detection, action recognition, semantic segmentation and object detection) by improving the design of and training existing machine learning components using Amazon Sagemaker
- Lead and supervise in-house data collection, curation and annotation efforts
- Document, patent and relay research findings to project stakeholders via presentations and reports

Key achievements:

- Reduced the cost of Just Walk Out technology by 8% by decreasing false negative action predictions on large product spaces in Amazon Fresh stores
- Reduced the cost of technology for detecting customer actions from streaming video by 6.5% through improvements to deep learning architectures and refined strategies for selecting system inputs
- Implemented and productionized an object detection system that achieves >95% mean average precision despite variations in background, lighting, appearance/packaging and occlusion
- Improved top-5 accuracy of in-house classification methods by 15% by improving training strategies for deep learning algorithms while respecting latency constraints
- Oversaw the creation of a massive real-world image dataset (>30k images) with image classification and bounding box labels for training in-house object detection systems

Graduate Research Assistant @ University of Illinois at Urbana-Champaign 01/16 – 08/21
Urbana, IL ***(5 years, 6 months)***

Responsibilities:

- Train and evaluate deep learning methods for object detection (RetinaNet, YOLOv3), object tracking (FCNT) and action recognition (SVM, HMM, MS-TCN, i3D) of construction workers and heavy machinery
- Collect research data using camera-equipped drones and build tools for crowdsourcing and monitoring large-scale ground truth annotation for image and video datasets

- Publish research findings in computer vision and construction journals and conference proceedings and present findings at conferences

Key achievements:

- Published 7 first-author, 3 second-author conference and journal papers, totalling 700+ citations
- Introduced the first annotated video dataset for end-to-end detection, tracking and activity recognition of earthmoving equipment, and a baseline averaging >85% action recognition accuracy across earthmoving activities
- Created an online tool for pose estimation annotation in videos and used it to coordinate the efforts of 71 annotators to create a dataset of 400 construction worker videos, which has been used in 5 of our lab's academic publications

Research Intern @ Autodesk 05/20 – 08/20
Toronto, ON, Canada ***(3 months)***

Responsibilities:

- Design, train and evaluate RNN-, LSTM- and VAE-based autoregressive generative models for 3D part hierarchies using PyTorch and compare them to the state-of-the-art

Key achievements:

- Published and presented research findings (ICCV 2021), in which we introduced the first generative model for 3D part hierarchies that can re-generate its outputs' parts

Education

Ph.D. in Computer Science @ University of Illinois at Urbana-Champaign 01/16 – 08/21
Urbana, IL ***(5 years, 7 months)***
Advisors: David Forsyth and Mani Golparvar-Fard
Thesis: Vision-based monitoring and design of built environments

MSc in Applied Mathematics @ Université Lille 1 09/14 – 09/15
Lille, France ***(1 year)***

BSc/MSc in Data Science/Machine Learning @ École Centrale de Lille 09/11 – 09/15
Lille, France ***(4 years)***

Selected Publications

[**ICCV 2021**] **D. Roberts**, A. Danielyan, H. Chu, M. Golparvar-Fard, D. Forsyth *LSD-StructureNet: Modeling Levels of Structural Detail in 3D Part Hierarchies*

[**Journal of Computing in Civil Engineering 2020**] W. Torres Calderon, **D. Roberts**, M. Golparvar-Fard *Synthesizing pose sequences from 3D assets for vision-based activity analysis*

[**Journal of Computing in Civil Engineering 2020**] **D. Roberts**, S. Tang, W. Torres Calderon, M. Golparvar-Fard *Vision-based construction worker activity analysis informed by body posture*

[**Automation in Construction 2020**] S. Tang, **D. Roberts**, M. Golparvar-Fard *Human-object interaction recognition for automatic construction site safety inspection*

[**ASCE International Conference on Computing in Civil Engineering 2019**] **D. Roberts**, M. Wang, A. Sabet, M. Golparvar-Fard *Annotating 2D imagery with 3D kinematically configurable assets of construction equipment for training pose-informed activity analysis and safety monitoring algorithms*

Technical Skills

Languages: Python (preferred), C/C++, MATLAB, JavaScript, Java, R, Swift
Libraries: PyTorch (preferred) + PyTorch Lightning, TensorFlow, NumPy, Pandas, Hugging Face
Other: GNU/Linux, Unity, Autodesk 360, Google Tango, ROS, SQL, HTML, CSS

Service Experience

Conference reviewer: CVPR 2021
Journal reviewer: Automation in Construction, Journal of Computing in Civil Engineering

Lead Teaching Assistant for CS 445, Computational Photography, University of Illinois at Urbana-Champaign, Spring 2021